Glossary

University mathematics has its own vocabulary, and much of it will be new to you. This glossary collects the words in this book that you are unlikely to have met at school, with a short explanation of each and, where it helps, a pointer to the section where the idea is used or explained properly. The explanations are meant to get you started, not to be precise definitions; where a word has a careful definition, the section it points to gives it.

A few words, such as *set*, *function* and *proof*, will be familiar, but are used at university with more care than at school, and they are included for that reason. Words in *italics* within an entry have entries of their own.

A

Adiabatic invariant A quantity that stays almost constant when a system is changed slowly. A molecule bouncing in a slowly compressed cylinder keeps its speed multiplied by the length of the cylinder nearly fixed (Section 10.4).

Algebraic number A number that is a root of a polynomial equation with integer coefficients (positive, negative or zero), not all zero. $\sqrt{2}$, i and $\frac{1}{2}(1+\sqrt{5})$ are algebraic; numbers that are not algebraic are *transcendental* (Section 6.3).

Analysis The branch of mathematics that puts calculus on a rigorous footing: limits, continuity, differentiation, integration and infinite series, all defined precisely and proved from first principles. Chapters 2 to 4 are an introduction to it.

Angular momentum The rotational counterpart of momentum. For a body spinning about an axis it is the *moment of inertia* multiplied by the angular velocity, and it is conserved when there is no turning effect from outside (Section 11.6).

Arithmetic–geometric mean The common limit of two sequences formed by repeatedly replacing a pair of numbers by their arithmetic mean and their geometric mean. It converges extraordinarily fast (Section 2.6).

Associative An operation $*$ is associative if $(a*b)*c = a*(b*c)$ for all a, b, c, so that brackets can be left out. Addition and multiplication of numbers are associative; subtraction is not.

Asymptotic Two expressions are asymptotic, written $f(n) \sim g(n)$, if their ratio tends to 1 as $n \to \infty$. Stirling's formula says that $n! \sim \sqrt{2\pi n}\,(n/\mathrm{e})^n$, even though the difference between the two sides grows without limit (Section 12.23).

Axiom A statement taken as a starting point, without proof, from which everything else

is deduced. Much of university algebra starts from a short list of axioms and asks what follows from them (Section 7.3, Section 6.5).

B

Base case The first step of a proof by *induction*: checking that the statement is true for the smallest value, usually $n = 0$ or $n = 1$. The *inductive step* then shows that truth for one value implies truth for the next.

Base-rate fallacy The mistake of judging the probability of something from the evidence alone while ignoring how common it was to begin with. If a disease is rare enough, even a positive result from an accurate test can be more likely to be a false alarm than not; whether it is depends on how rare the disease is and how accurate the test is (Section 12.2).

Basis A set of vectors such that every vector in the space can be written in exactly one way as a combination of them. In the plane, $\mathbf{i}$ and $\mathbf{j}$ form a basis; so does any pair of vectors that are not parallel. The number of vectors in a basis is the *dimension* of the space (Section 6.3).

Bayes' theorem The rule for turning one conditional probability round: $\mathrm{P}(A \mid B) = \mathrm{P}(B \mid A)\,\mathrm{P}(A)/\mathrm{P}(B)$. It tells you how to update a belief in the light of evidence (Section 12.2).

Bias An *estimator* is biased if, on average over many samples, it comes out too high or too low; it is *unbiased* if its expectation is exactly the quantity it estimates (Section 13.3).

Bijection A one-to-one correspondence: a rule that pairs off every member of one set with exactly one member of another, with nothing left over on either side. Two finite sets have the same size exactly when there is a bijection between them, and this is how the size of an infinite set is defined (Section 1.8, Section 9.10).

Binary operation A rule for combining two objects to give a third, such as addition of numbers, multiplication of matrices, or composition of functions.

Bound An upper bound for a set or sequence of numbers is a number that none of them exceeds; a lower bound is one that none of them falls below. A sequence is *bounded* if it has both (Section 2.2).

Branching process A model of a population in which each individual independently has a random number of offspring. The central question is whether the family line dies out, and the answer comes from a *probability generating function* (Section 12.15).

C

Calculus of variations The study of problems in which the unknown is a whole function or curve, chosen to make some quantity as small or as large as possible: the shortest path, the quickest slide, the curve enclosing the most area (Section 4.7).

Catenary The shape of a hanging chain, the graph of $y = a\cosh(x/a)$. It looks like a parabola but is not one (Section 11.14).

Cauchy distribution A continuous distribution, with a bell-shaped graph, whose tails are so heavy that it has no mean. The average of a sample from it is no more reliable than a single observation (Section 12.22).

Central limit theorem The theorem that the sum of a large number of independent random quantities with the same distribution and finite variance, suitably shifted and scaled, has approximately a normal distribution, whatever that common distribution is. Versions exist for quantities with different distributions, provided no few of them dominate the sum. It explains why the bell curve appears everywhere (Section 12.24).

Chaos Behaviour of a deterministic system that is so sensitive to its starting point that long-term prediction is impossible in practice: two starting points that are very close together lead to completely different futures (Section 8.7, Section 8.8).

Closed A set is closed under an operation if combining two members of the set always gives another member. The even numbers are closed under addition; the odd numbers are not.

Commutative An operation $*$ is commutative if $a * b = b * a$ always. Multiplication of numbers is commutative; multiplication of matrices and of *quaternions* is not (Section 7.1, Section 6.7).

Completeness The property of the real numbers that they have no gaps: every increasing sequence that is bounded above converges to a real number. The rational numbers lack it, since a sequence of fractions can close in on $\sqrt{2}$. Much of *analysis* rests on it (Section 2.2).

Congruence $a \equiv b \pmod{n}$, read "a is congruent to b modulo n", means that a and b leave the same remainder on division by n; equivalently, n divides $a - b$. Arithmetic with congruences is called *modular arithmetic* (Section 1.5).

Conjecture A statement that is believed to be true but has not been proved.

Conservation law A statement that some quantity, such as energy, momentum or angular momentum, does not change as a system evolves. Conservation laws often let you solve a problem without following its details (Section 10.1).

Continuous A function is continuous if small changes in the input produce small changes in the output; informally, its graph can be drawn without lifting the pen. Precisely, f is continuous at a if $f(x) \to f(a)$ as $x \to a$ (Section 3.1, Section 8.8).

Contradiction, proof by A proof that assumes the statement is false and deduces something impossible. The classic example is the proof that $\sqrt{2}$ is irrational (Section 1.3).

Contrapositive The contrapositive of "if P then Q" is "if not Q then not P". The two statements are logically equivalent, and proving the contrapositive is often easier. It is not the same as the *converse*, "if Q then P", which may well be false.

Converge A sequence converges if its terms get, and stay, as close as you like to a fixed number, its *limit*. A series converges if its *partial sums* converge. A sequence or series that does not converge *diverges* (Section 2.1, Section 2.3).

Convex A function is convex if the chord joining any two points on its graph lies on or above the graph, as for x^2 or e^x. Many inequalities, including AM–GM, are consequences of convexity (Section 3.3).

Corollary A result that follows quickly from a *theorem* just proved.

Correlation A number between -1 and 1 measuring how close two random quantities are to having a straight-line relationship. It is the *covariance* scaled by the two standard deviations (Section 13.7).

Countable A set is countable if its members can be listed as a first, second, third, and so on, possibly for ever. The whole numbers and, surprisingly, the fractions are countable; the real numbers are not (Section 1.8).

Counterexample A single example showing that a general statement is false. One counterexample is enough to disprove a claim, however many examples support it.

Covariance $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$, a measure of how two random quantities vary together. It is zero for independent quantities, though zero covariance does not imply independence (Section 12.4, Section 13.7).

Cycle A set of points that a repeated function visits in turn: $x_1 \mapsto x_2 \mapsto \cdots \mapsto x_n \mapsto x_1$. The number n is its *period*, and the points are *periodic points* (Section 8.8).

Cycloid The curve traced by a point on the rim of a rolling wheel. Turned upside down, it is both the curve of quickest descent and the curve along which every slide takes the same time (Section 11.4).

Cyclotomic polynomial The polynomial $\Phi_n(x)$ whose roots are exactly the *primitive* nth *roots of unity*. It has integer coefficients and is *irreducible*, and $x^n - 1$ is the product of the $\Phi_d(x)$ for all divisors d of n: for example, $x^4 - 1 = (x-1)(x+1)(x^2+1)$ (Section 6.1).

D

Derangement A rearrangement of a list in which nothing stays in its original position (Section 9.8).

Derivation An operation D on functions (or on other objects) with $D(f+g) = Df + Dg$ and the product rule $D(fg) = f\,Dg + g\,Df$. Differentiation is the standard example, and much of its behaviour follows from these two rules alone (Section 6.5).

Determinant A number calculated from a square matrix, which measures the factor by which the corresponding transformation scales areas or volumes. It is zero exactly when the matrix has no inverse (Section 7.3).

Differentiable A function is differentiable at a point if its graph has a well-defined tangent there, so that the limit defining the derivative exists. Every differentiable function is *continuous*, but not the other way round: $|x|$ is continuous but not differentiable at 0.

Dimension The number of vectors in a *basis*: the number of independent directions in a *vector space*.

Diverge See *converge*.

Dynamical system A rule describing how the state of a system changes with time, either step by step (a sequence defined by $x_{n+1} = f(x_n)$) or continuously (a differential equation). Chapter 8 is about them.

E

Eigenvalue, eigenvector A non-zero vector $\mathbf{v}$ is an eigenvector of a matrix $\mathbf{M}$ if $\mathbf{Mv} = \lambda\mathbf{v}$ for some number λ, its eigenvalue: the transformation stretches $\mathbf{v}$ without turning it (Section 7.2).

Envelope A curve that touches every member of a family of curves. The envelope of all the trajectories of a ball thrown at a fixed speed is the *safety parabola*, the boundary of the region the ball can reach (Section 11.9).

Equilibrium A state of a system that does not change. It is *stable* if small disturbances stay small, and *asymptotically stable* if they also die away, as they do for a pendulum with friction; a frictionless pendulum hanging straight down is stable, but a small push makes it swing for ever. It is *unstable* if some small disturbances grow (Section 8.3, Section 11.8).

Estimator A rule for calculating, from data, a guess at an unknown quantity, such as the size of a population or the mean of a distribution (Section 13.3).

Euler characteristic For a polyhedron with V vertices, E edges and F faces, the number $V - E + F$, which is always 2 for a polyhedron without holes (Section 7.9).

Existence proof A proof that something exists that does not necessarily say how to find it. The *pigeonhole principle* and the *intermediate value theorem* both give existence proofs.

Expectation The mean value of a random quantity X, written $\mathrm{E}(X)$: the long-run average of many independent repetitions. Expectation is additive, $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, whether or not X and Y are independent (Section 9.11).

Extremal principle The strategy of looking at the smallest, largest, first or last object of some kind, and showing that it has a property that leads either to the result or to a contradiction (Section 1.3).

F

Field A number system in which you can add, subtract, multiply and divide (except by zero), with all the usual rules holding. The rational, real and complex numbers are fields; so are the integers modulo a prime. The integers are not, because you cannot always divide (Section 6.3).

Field extension A larger field containing a smaller one, such as the numbers $a + b\sqrt{2}$ with a, b rational, which contain the rationals. Its *degree* is its *dimension* as a *vector space* over the smaller field (Section 6.3).

Fixed point A point x with $\mathrm{f}(x) = x$, which the function leaves where it is. On a graph, fixed points are where $y = \mathrm{f}(x)$ meets $y = x$ (Section 2.2, Section 8.8).

Frame of reference A point of view from which positions and motions are measured. An *inertial* frame is one in which Newton's first law holds; in an accelerating frame, such as a train speeding up, there appear to be extra forces (Section 11.5).

Function A rule that assigns to each member of one set, the *domain*, exactly one member of another set, the *codomain*. At university a function is not the same as a formula: many important functions have no formula, and some formulae do not define functions.

Functional equation An equation whose unknown is a function, such as $\mathrm{f}(x + y) = \mathrm{f}(x)\mathrm{f}(y)$ (Section 3.8).

Fundamental theorem of algebra Every polynomial of degree at least 1 with complex coefficients has a complex root, and so factorises completely into linear factors over the complex numbers.

G

Gamma function A function Γ that extends the factorial to non-integers: $\Gamma(n+1) = n!$ for whole numbers n, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, which in a sense says that $(-\frac{1}{2})! = \sqrt{\pi}$ (Section 4.4).

Gaussian integer A complex number $a + b\mathrm{i}$ with a and b integers. They behave much like ordinary integers, with their own primes and unique factorisation (Section 1.6).

Generating function A power series $a_0 + a_1 x + a_2 x^2 + \cdots$ whose coefficients are the terms of a sequence. Operations on the series, such as multiplication, correspond to useful operations on the sequence (Section 9.4). A *probability generating function* does this for the probabilities of a random whole number (Section 9.6, Section 12.15); a *moment generating function* is a close relative (Section 12.24).

Group A set with a *binary operation* that is *associative*, has an *identity* element and gives every element an *inverse*. The symmetries of a shape form a group, as do the non-zero complex numbers under multiplication. Groups are the language of symmetry (Section 6.6, Section 6.7).

H

Hyperbolic geometry A geometry in which, through a point not on a given line, there are infinitely many lines that never meet it, and the angles of a triangle add up to less than π. A disc with a suitable way of measuring distance is a model of it (Section 5.8).

I

Identity An identity element for an operation $*$ is an element e with $e * a = a * e = a$ for every a: 0 for addition, 1 for multiplication, the identity matrix $\mathbf{I}$ for matrix multiplication. Separately, an identity is an equation that holds for every value of its variables, such as $\cos^2\theta + \sin^2\theta \equiv 1$.

If and only if "P if and only if Q", often written "P iff Q" or $P \iff Q$, means that P and Q are either both true or both false. Proving it needs two arguments, one in each direction.

Independent Two events are independent if knowing that one has happened does not change the probability of the other: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$. Three or more events can be independent in pairs without being independent all together (Section 12.4).

Induction A method of proving a statement for every whole number n: prove it for the first value (the *base case*), then prove that whenever it is true for one value it is true for the next (the *inductive step*). In *strong induction*, the inductive step may use the statement for all smaller values, not just the previous one (Section 1.2).

Infinite descent A form of proof by contradiction: show that any solution of a problem in whole numbers would lead to a smaller one, and so to an endless decreasing sequence of positive whole numbers, which is impossible (Section 1.5).

Intermediate value theorem If a *continuous* function takes the values a and b, it takes every value in between. So a continuous function that changes sign must have a root (Section 8.8).

Invariant A quantity that does not change while a process runs, or under some transformation. Finding an invariant is one of the most powerful ways to prove that something is impossible, or to solve a problem without following every step (Section 7.7, Section 10.4).

Inverse An inverse of an element a, for an operation with *identity* e, is an element b with $a * b = b * a = e$. An inverse function undoes a function: $\mathrm{f}^{-1}(\mathrm{f}(x)) = x$.

Irrational A real number that cannot be written as a fraction $\frac{p}{q}$ with p and q integers ($q \neq 0$), such as $\sqrt{2}$, π or e (Section 1.4).

Irreducible A polynomial is irreducible if it cannot be written as the product of two polynomials of lower degree with coefficients of the same kind. $x^2 - 2$ is irreducible over the rationals, but factorises as $(x - \sqrt{2})(x + \sqrt{2})$ over the reals (Section 6.1).

Isomorphic Two structures, such as two *groups*, are isomorphic if there is a *bijection* between them that matches their operations exactly: they are the same structure with different labels (Section 6.7, Section 6.8).

Iteration Applying the same function again and again, so that $x_1 = \mathrm{f}(x_0)$, $x_2 = \mathrm{f}(x_1)$, and so on. The sequence x_0, x_1, x_2, $\ldots$ is the *orbit* of x_0 (Section 8.7).

L

Law of large numbers The theorem that the average of many independent repetitions of a random quantity tends to its *expectation*, provided the expectation exists (Section 12.22).

Lemma A result proved mainly as a step towards a larger *theorem*.

Limit The value that a sequence or function gets, and stays, arbitrarily close to. Making this idea precise was one of the great achievements of nineteenth-century mathematics, and it is the starting point of *analysis* (Section 2.1).

Limit cycle A closed loop in the motion of a system that nearby motions spiral towards (or away from), such as the steady rhythm of a heartbeat or a clock (Section 8.4).

Linear map A function T between vector spaces with $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(\lambda\mathbf{u}) = \lambda T(\mathbf{u})$. Every linear map can be represented by a matrix, and linear algebra is the study of them (Section 7.2).

Logistic map The rule $u_{n+1} = ru_n(1 - u_n)$, a simple model of a population limited by its food supply, and the standard example of *chaos* (Section 8.7).

M

Markov chain A random process that moves between states in steps, in which the probability of the next step depends only on the present state, not on how the process got there (Section 12.16).

Maximum likelihood A way of estimating an unknown quantity: choose the value that makes the observed data most probable (Section 13.2).

Memoryless A random waiting time is memoryless if, however long you have already waited, the remaining wait has the same distribution as it did at the start. The exponential distribution is the only continuous memoryless distribution (Section 12.18).

Minimal polynomial The polynomial of lowest degree, with rational coefficients and leading coefficient 1, that has a given algebraic number as a root. The minimal polynomial of $\sqrt{2}$ is $x^2 - 2$ (Section 6.3).

Modular arithmetic See *congruence*.

Moment of inertia The rotational counterpart of mass: a measure of how hard it is to change a body's rate of rotation about an axis. It depends on how far the mass is

from the axis, not just on how much mass there is (Section 11.7).

Monotonic A sequence or function is monotonic if it never goes down (increasing) or never goes up (decreasing). A monotonic sequence that is bounded must converge (Section 2.2, Section 11.10).

N

Necessary and sufficient P is a necessary condition for Q if Q cannot happen without P ($Q \Rightarrow P$); it is sufficient if P guarantees Q ($P \Rightarrow Q$). A condition that is both is equivalent to Q.

Normal modes The special patterns of vibration of a system in which every part moves in step at the same frequency. Any vibration of the system is a combination of its normal modes (Section 11.2).

O

Order The order of an element g of a *group* is the smallest positive whole number n for which g^n is the *identity*: the number of times you must apply it to get back to the start (Section 7.4). If there is no such n, as for the number 1 in the integers under addition, g has *infinite order*. The order of a finite group is the number of elements it has.

Orthogonal Perpendicular. At university the word is extended beyond vectors: two functions f and g can be called orthogonal if $\int \mathrm{f}(x)\mathrm{g}(x)\,\mathrm{d}x = 0$ over a suitable range (Section 8.6).

P

Parameter A quantity that is fixed within a particular problem but can vary from one problem to another, such as r in the *logistic map*. Many STEP questions ask how the answer depends on a parameter.

Partial sum The sum $a_1 + a_2 + \cdots + a_n$ of the first n terms of a series. The series *converges* if its partial sums do (Section 2.3).

Pigeonhole principle If more than n objects are put into n boxes, some box contains at least two of them. Simple as it is, it proves some remarkable *existence* results (Section 9.2).

Poisson process A model of events that occur at random, independently, at a constant average rate, in time or in space: radioactive decays, phone calls, stars in the sky (Section 12.19, Section 12.18).

Primitive root of unity An nth root of unity whose powers give all n of the nth roots of unity; equivalently, one that is not an mth root of unity for any smaller m (Section 6.1).

Probability density function For a continuous random quantity, the function f whose integral over an interval gives the probability of landing in that interval.

Proof A complete logical argument that a statement is true, starting from definitions, axioms and results already proved. At university, a calculation that gives the right answer is not a proof unless each step is justified.

Q

Quaternions A number system discovered by William Rowan Hamilton in 1843, extending the complex numbers with three square roots of -1, written i, j, k, which satisfy $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1$. Their multiplication is not *commutative*. They are used to describe rotations in three dimensions (Section 6.7, Section 7.5).

R

Random variable A quantity whose value is determined by the outcome of a random experiment, such as the number of heads in ten tosses of a coin.

Random walk A path made of successive random steps, such as a counter that moves one place left or right on the toss of a coin (Section 9.10, Section 12.17).

Rapidity A measure of speed in special relativity that, unlike velocity, simply adds when one motion is combined with another in the same direction. If c is the speed of light, velocity v has rapidity $\operatorname{artanh}(v/c)$ (Section 7.6).

Recurrence relation An equation giving each term of a sequence in terms of earlier ones, such as $F_{n+1} = F_n + F_{n-1}$ for the Fibonacci numbers.

Resonance The large response of a system that is pushed periodically at, or near, its own natural frequency, as when a child on a swing is pushed in time with the swing (Section 11.1).

Riemann integral The precise definition of the integral as the limit of sums of areas of thin rectangles, when that limit exists (Section 4.1).

Rigour The standard of argument in which every step is justified and nothing is taken for granted because it looks obvious or is true in the examples you have tried (Chapter 3).

Root of unity A complex number z with $z^n = 1$ for some positive whole number n. The nth roots of unity are $\mathrm{e}^{2\pi \mathrm{i} k/n}$ for $k = 0, 1, \ldots, n-1$, which form a regular n-gon on the unit circle (Section 5.2).

S

Sample space The set of all possible outcomes of a random experiment. Events are subsets of it (Section 12.1).

Set A collection of objects, its *elements*, considered as a single thing. The order of the elements does not matter, and nor does repetition: $\{1,2,3\}$ and $\{3,1,2,2\}$ are the same set.

Simple harmonic motion Motion in which the acceleration is proportional to the displacement from a fixed point and directed towards it, $\ddot{x} = -\omega^2 x$. The solution is $x = A\cos(\omega t + \varepsilon)$, and the period, $2\pi/\omega$, does not depend on the size of the swing (Section 11.4).

Stable, unstable See *equilibrium*.

Stirling's formula The approximation $n! \approx \sqrt{2\pi n}\,(n/\mathrm{e})^n$, whose percentage error tends to zero as n grows (Section 12.23).

Subgroup A subset of a *group* that is itself a group under the same operation (Section 6.6).

Symmetry argument An argument that two quantities must be equal because nothing in the problem distinguishes between them. A good symmetry argument can replace pages of calculation (Section 12.5).

T

Theorem An important statement that has been proved. Lesser results are often called *propositions*, those proved on the way to a theorem *lemmas*, and quick consequences *corollaries*.

Trace The sum of the entries on the leading diagonal of a square matrix. Like the *determinant*, it does not change when the matrix is written in a different *basis*, and it equals the sum of the *eigenvalues* (Section 7.4, Section 7.7).

Transcendental A number that is not *algebraic*: it is not the root of any polynomial equation with integer coefficients, not all zero. e and π are transcendental, though proving it is hard (Section 1.4).

Triangle inequality $|a+b| \leq |a| + |b|$, for real numbers, complex numbers or vectors: one side of a triangle is never longer than the other two together.

U

Unique factorisation The property that every whole number greater than 1 can be written as a product of primes in only one way, apart from the order of the factors. It is not as obvious as it looks: there are number systems in which it fails (Section 1.7).

Uniqueness theorem A theorem saying that a problem has only one solution. For differential equations, uniqueness can fail when the equation is not well behaved, so that a single starting point leads to several different futures (Section 8.1).

V

Vector space A collection of objects, called vectors, that can be added together and multiplied by numbers, following the familiar rules for arrows in the plane. Polynomials, functions and solutions of many differential equations all form vector spaces, so the methods of linear algebra apply to them (Section 6.3).

W

Well-defined An object or rule is well-defined if its definition really does specify a single unambiguous thing: if it does not depend, for example, on an arbitrary choice made along the way.

Well-ordering principle Every non-empty set of positive whole numbers has a smallest member. It is equivalent to the principle of *induction*, and it lies behind every argument that begins "take the smallest counterexample" (Section 1.3).

Without loss of generality A phrase, often shortened to WLOG, meaning that the argument considers only one case because the other cases are the same apart from relabelling. "Without loss of generality, $a \leq b$" means that if $a > b$ you can swap their names.